

Mode of Examination: Online
M.Sc Semester – I Mid Semester Examination, 2022
Subject : Object Oriented Analysis and Design (CSMC104)

Full Marks: 30

Date: 03.02.2022

Time and Duration: 1½ hour

Please note the following instructions carefully:

Promise not to commit any academic dishonesty.

Marks will be deducted if the same/similar answers are found in different answer-scripts.

Candidates are required to answer in their own words as far as applicable.

Each page of the answer scripts should have your University Roll # on the right-top corner.

The name of the scanned copy of the answer script will be of the following format:

(Example: CSMC104-OOAD-MidSem-My Roll Number.pdf)

The subject of the mail should be the file name only.

The name of the scanned answer-script is to be sent to **cucse2020@gmail.com**

The report should have the top page (Page #1) as an index page; mention page number(s) against the answer of each question number.

The answer-script may not be accepted after the scheduled time.

1. Answer any five questions: [5 x 2]
 - a. List different modifiers used in Java.
 - b. How many String objects are created by the following java code?

```
String s1 = "abc";  
String s2 = new String("abc");  
String s3 = new String("efg");
```
 - c. What is the basic difference between Demon Thread and Non Demon Thread? Give one example of each.
 - d. Write a small code snippet to create an object of the class "com.cu.Xyz" without using new operator in java.
 - e. Critically comment on the statement – "Java is both compiled and interpreted".
 - f. We want to open a file as soon as the Class is loaded and close the file before the corresponding object is removed from the heap. Name the features to be used in Java to implement it.
2. Answer any four questions: [4 x 5]
 - a. Differentiate between coupling and cohesion. What is the relation between abstraction, information hiding and encapsulation? Explain with proper example.
 - b. Differentiate between IS-A and HAS-A relation between two objects. How can we reuse code and reuse design? Critically comment on the statement – "Multiple Inheritance is loosely implemented in Java".
 - c. Compare and contrast dynamic binding and static binding with suitable examples.

- d. The logs of a system are stored in a stable storage like hard disk. Design a class called “Logger” that stores the logs in an array of String during initialization of a Logger object. Different modules of the system will use the Logger object to read and write different logs generated in the system. Apply appropriate design pattern for efficient implementation of the Logger class.
- e. An Employee Management System (EMS) manages different types of employees namely CEO, Manager, Team Lead and Worker. Workers are subordinates of a Team Lead while Team Leads work under a Manager and Managers work under the CEO. Design appropriate classes so that EMS can easily handle joining or leaving of an employee under or from a supervisor and can return number of employees work under a particular employee.
